Day – 6 Layer Switch Inter-VLAN Routing Theory.

Task – 1 Create three VLANs.

- Step 1: Create the VLANs on the Layer 3 Switch
- Open your Layer 3 switch CLI and enter global configuration mode to create the three VLANs:
- Switch> enable
- Switch# configure terminal
- ! Create Sales VLAN
- Switch(config)# vlan 10
- Switch(config-vlan)# name Sales
- Switch(config-vlan)# exit
- ! Create Support VLAN
- Switch(config)# vlan 20
- Switch(config-vlan)# name Support
- Switch(config-vlan)# exit
- ! Create Marketing VLAN
- Switch(config)# vlan 30
- Switch(config-vlan)# name Marketing
- Switch(config-vlan)# exit
- Step 2: Assign Access Ports to the VLANs
- Map individual physical interfaces to your newly created VLANs so that PCs connected to those ports belong to the correct broadcast domain:
- ! Assign FastEthernet 0/1 to Sales (VLAN 10)
- Switch(config)# interface fastEthernet 0/1
- Switch(config-if)# switchport mode access
- Switch(config-if)# switchport access vlan 10
- Switch(config-if)# exit
- ! Assign FastEthernet 0/2 to Support (VLAN 20)
- Switch(config)# interface fastEthernet 0/2
- Switch(config-if)# switchport mode access
- Switch(config-if)# switchport access vlan 20
- Switch(config-if)# exit
- ! Assign FastEthernet 0/3 to Marketing (VLAN 30)
- Switch(config)# interface fastEthernet 0/3
- Switch(config-if)# switchport mode access
- Switch(config-if)# switchport access vlan 30
- Switch(config-if)# exit

- Step 3: Verify the VLAN Configuration
- To confirm that all three VLANs are created and their respective ports are assigned properly, run the verification command:
- Switch# show vlan brief
- Expected Output:
- You should see a table displaying your custom VLAN names alongside their assigned switch ports:
- VLAN Name Status Ports
- ---- --------------- --------- -------------------------------
- 1 default active Fa0/4, Fa0/5, Fa0/6, ...
- 10 Sales active Fa0/1
- 20 Support active Fa0/2
- 30 Marketing active Fa0/3

Task – 2 Config switch virtual interfaces to each VLAN.

- Step 1: Enable Routing on the Layer 3 Switch
- Before configuring SVIs, you must globally enable IP routing on the Layer 3 switch so it can route packets between different VLANs:
- Switch> enable
- Switch# configure terminal
- Switch(config)# ip routing
- Step 2: Configure the Switch Virtual Interfaces (SVIs)
- Create an SVI for each VLAN and assign its respective default gateway IP address:
- ! Configure SVI for VLAN 10 (Sales)
- Switch(config)# interface vlan 10
- Switch(config-if)# ip address 192.168.10.1 255.255.255.0
- Switch(config-if)# no shutdown
- Switch(config-if)# exit
- ! Configure SVI for VLAN 20 (Support)
- Switch(config)# interface vlan 20
- Switch(config-if)# ip address 192.168.20.1 255.255.255.0
- Switch(config-if)# no shutdown
- Switch(config-if)# exit
- ! Configure SVI for VLAN 30 (Marketing)
- Switch(config)# interface vlan 30
- Switch(config-if)# ip address 192.168.30.1 255.255.255.0
- Switch(config-if)# no shutdown
- Switch(config-if)# exit

- Step 3: Verify SVI Status
- To confirm that the SVIs are active and in the up/up state, run the following verification command:
- Switch# show ip interface brief
- Expected Output:
- You should see your VLAN interfaces listed with their assigned IP addresses and an operational status of up/up:
- Interface IP-Address OK? Method Status Protocol
- Vlan10 192.168.10.1 YES manual up up
- Vlan20 192.168.20.1 YES manual up up
- Vlan30 192.168.30.1 YES manual up up

Task – 3 Enable IP routing on the layer 3 switch.

- To enable the Layer 3 switch to forward traffic between your VLANs internally without needing an external router, run the following command in global configuration mode:
- Switch> enable
- Switch# configure terminal
- Switch(config)# ip routing
- Switch(config)# end
- Switch# write memory

Task – 4 Verify inter-VLAN routing by connecting two pcs.

- Step 1: Connecting and Configuring the PCs
- To test the Layer 3 switch routing configuration, set up your end devices with the following IP addresses and default gateways:
- PC 0 (Sales / VLAN 10):
- IP Address: 192.168.10.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1 (Points to Vlan10 SVI)
- *Connected to Switch Port:* FastEthernet 0/1
- PC 1 (Support / VLAN 20):
- IP Address: 192.168.20.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.20.1 (Points to Vlan20 SVI)
- *Connected to Switch Port:* FastEthernet 0/2

- Step 2: Executing the Ping Test
- Open PC 0 in Packet Tracer.
- Navigate to Desktop > Command Prompt.
- Ping PC 1 across the VLAN boundary:
- C:\> ping 192.168.20.10
- Step 3: Captured Ping Results
- Pinging 192.168.20.10 with 32 bytes of data:
- Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
- Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
- Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
- Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
- Ping statistics for 192.168.20.10:
- Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
- Approximate round trip times in milliseconds:
- Minimum = 0ms, Maximum = 0ms, Average = 0ms

Task – 5 Refactor your VLAN and SVI config.

- Step 1: Updating the SVI Configuration for VLAN 30
- To change VLAN 30 (Marketing) from a /24 (255.255.255.0) subnet mask to a /25 (255.255.255.128) subnet mask, open the Layer 3 switch CLI and reconfigure the SVI:
- Switch> enable
- Switch# configure terminal
- ! Reconfigure SVI for VLAN 30 with a /25 subnet mask
- Switch(config)# interface vlan 30
- Switch(config-if)# ip address 192.168.30.1 255.255.255.128
- Switch(config-if)# exit
- Switch(config)# end
- Switch# write memory
- Step 2: Updating the PC Configuration in VLAN 30
- Any PC assigned to VLAN 30 must also be updated to match the new subnet parameters:
- IP Address: 192.168.30.10
- Subnet Mask: 255.255.255.128 (spanning range .1 to .127)
- Default Gateway: 192.168.30.1
- Step 3: What Changes Were Made?

- Subnet Range Reduction: A /25 mask divides the 192.168.30.0 network in half, providing 128 total IP addresses (.0 to .127) instead of the 256 addresses provided by a /24 mask.
- Gateway and Usable Hosts: The SVI address remains 192.168.30.1, but usable host IPs in this subnet are now restricted to 192.168.30.2 through 192.168.30.126 (with .127 acting as the local broadcast address).
- Step 4: How This Affects Communication with Other VLANs
- Inter-VLAN Routing Still Works: Communication between VLAN 30 and other VLANs (like Sales on VLAN 10 or Support on VLAN 20) continues to function normally. Because routing happens at Layer 3 via the switch's internal routing table, the switch successfully routes packets between different subnets regardless of whether they use /24, /25, or mixed subnet masks.
- Subnet Boundary Enforcement: Any device in VLAN 30 configured with an IP address above 192.168.30.127 (for example, 192.168.30.130) will now be considered outside the local subnet. Without an explicit router or secondary IP configuration, those high-range devices would fail to communicate properly because they would no longer be recognized as part of the direct local broadcast domain.